

cardiac massage being instituted much more frequently than formerly; consequently cases of residual cerebral hypoxic damage may be encountered in increasing numbers. Some of the cerebral damage in these cases may be irreversible, but probably a larger proportion of cells are reversibly damaged. It does not seem to be sufficiently realized that in addition to nerve cell injury there is cerebral vascular damage due to the acute hypoxic episode, and increased permeability from damaged capillaries leads to subsequent cerebral oedema. This oedema fluid acts as a physical barrier and cell respiration is impaired; in addition C.S.F. pressure is probably raised and venous obstruction encouraged, all of which mitigate against cellular recovery and reversibly damaged cells now succumb.

Sadove, Wyant, and Gittelson¹ recommended that cases of cardiac arrest should be treated by a dehydration regimen to combat this cerebral oedema. We have reported some cases of severe cerebral hypoxia in which dehydration with I.V. 50% sucrose was instituted² and were favourably impressed with the results; other cases since then have confirmed our views. Experimental work on dogs on which we are now engaged at the Clinical Research Institute of the Middlesex Hospital also encourages us as to its efficacy.

Cerebral hypoxia with residual neurological lesions can follow hypoxic episodes from many causes less severe than cardiac arrest, and we believe that, if dehydration therapy were instituted, recovery would be more frequent and certainly more rapid.—We are, etc.,

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REFERENCES

- ¹ Sadove, M. S., Wyant, G. M., and Gittelson, L. A., *Brit. med. J.*, 1953, 2, 255.
- ² Argent, D. E., and Cope, D. H. P., *ibid.*, 1956, 1, 593.

Neurological Complications of Pertussis Immunization

SIR,—Dr. J. M. Berg's timely review of the neurological complications of pertussis immunization (*Journal*, July 5, p. 24) has prompted us to record our own experience of similar cases. In the past 10 years one of us (D.C.T.-P.) has seen six patients with neurological reactions developing within 24 hours of pertussis immunization. One of these cases was fatal and three were left with permanent brain damage.

Case 1.—A male infant, whose development had been normal in every way, was quite well until he had convulsions nine hours after his third inoculation with combined diphtheria and whooping-cough vaccine. During the attack he was unconscious for 10 hours with right-sided convulsions, which were followed by a temporary right-sided hemiplegia. He continued to have major fits of a similar nature periodically and also attacks of petit mal. His behaviour was later characterized by marked restlessness and overactivity, and he showed all the signs of the "hyperkinetic syndrome," described by Ounsted.¹ His mental development, which appeared to be well advanced up to his immunization at 9 months, became retarded.

Case 2.—A male child, who had been quite healthy since birth, received his first dose of pertussis vaccine at the age of 8 months, at the same time as his twin brother. Seven hours later he developed convulsions, which continued intermittently until he was admitted to hospital, comatose, in status epilepticus. He died soon after admission, 14 hours after being immunized. Necropsy was performed nine hours after death. The only relevant findings were cerebral oedema and intense congestion. Microscopic examination failed to show any evidence of haemorrhage, encephalitis, or degeneration of neurons, as described by Globus and Kohn.² The other twin remained unaffected, and there was no family history of epilepsy. A report of the coroner's inquest in this case appeared in the medico-legal column of this *Journal*.³

Case 3.—A male infant, who had had two convulsions at the age of 6 months, made normal progress until his second combined diphtheria and whooping-cough inoculation at the age of 13 months. About 12 hours after the injection he developed status epilepticus, followed by a left hemiplegia; the paresis gradually improved, leaving a residual weakness of the left hand. He continued to suffer from periodic major and minor fits and his mental development became retarded. His behaviour pattern was also hyperkinetic, and at intervals he was intensely restless, over-

active, and destructive. The similarity of behaviour to that in our first patient was most striking. When last seen, at the age of 4, his gait was still ataxic.

Case 4.—A male infant aged 5 months developed convulsions and prolonged unconsciousness seven hours after his first inoculation with combined diphtheria and whooping-cough vaccine. He had a past history of one convulsion, lasting 10 minutes, five weeks previously. Until he was immunized his progress had been, if anything, in advance of normal, for he sat up at the age of 4½ months. He continued to have fits periodically. When he was learning to walk at the age of 1, he was found to be very ataxic, and two years later he was still unable to walk alone. His behaviour was restless and hyperkinetic, though this was not so marked and troublesome as in Cases 1 and 3.

Case 5.—A male child aged 6 months had been well until his second injection of combined diphtheria and whooping-cough vaccine. Following the inoculation he slept for three and a half hours and then woke with a persistent unnatural cry. He vomited once, but there were no convulsions. During the next two weeks he remained miserable, listless, and made no attempt to sit up as before. His condition was described by his parents as "completely lifeless." He then recovered, apart from some loss of weight and appetite during the next fortnight. He appeared quite normal when he was last seen four months later.

Case 6.—A normal healthy male infant aged 6 months some hours after receiving his first dose of combined diphtheria and whooping-cough vaccine, was noticed by his mother to be abnormally limp and drowsy. He lay in a stuporose condition for about 12 hours, frequently emitting shrill whining cries. There was no pyrexia or evidence of infection. The child then made a gradual complete recovery. This was a relatively mild reaction, which is quite possibly not uncommon. It is included as it may be an example of the type of case which would sustain severe brain damage if further doses of pertussis vaccine had been given.

The fact that six cases of neurological complications following the administration of pertussis vaccine have occurred in our limited experience suggests that the danger of whooping-cough immunization are greater than Dr. Berg and others seem to think. Although whooping-cough with all its attendant complications constitutes a serious risk in young children, the disease seldom produces permanent brain damage. It is therefore very doubtful whether the natural disease does in fact afford a greater danger of neurological sequelae than prophylactic immunization.

The significance of recent pertussis immunization was overlooked in two of our patients when they were first admitted with convulsions. It is possible, therefore, that on many occasions encephalopathy following pertussis immunization is not diagnosed, especially in the milder cases. The hyperkinetic syndrome is usually associated with epilepsy and brain damage arising from an organic cerebral insult in early life. Encephalopathy due to pertussis immunization, so far as we are aware, has not been previously reported as a cause of this syndrome. It would be interesting to learn whether others have encountered similar cases.

We agree with Dr. Berg that pertussis immunization is contraindicated in all children with a history of convulsions and also in those who have had minor reactions to the initial dose of the vaccine.—We are, etc.,

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REFERENCES

- ¹ Ounsted, C., *Lancet*, 1955, 2, 303.
- ² Globus, J. H., and Kohn, J. L., *J. Amer. med. Ass.*, 1949, 141, 507.
- ³ *Brit. med. J.*, 1949, 2, 1478.

Medical War Crimes

SIR,—Several of your correspondents have expressed their dismay at the fact that Dr. Herta Oberheuser, convicted of medical war crimes, has been permitted to resume practice in Germany. It is good to know that the British Medical Association have now formally protested against this cynical decision. It is also good to know that the West German Medical Association and the Ministry of the Interior of Schleswig-Holstein have instituted proceedings for the withdrawal of her approbation to practise.